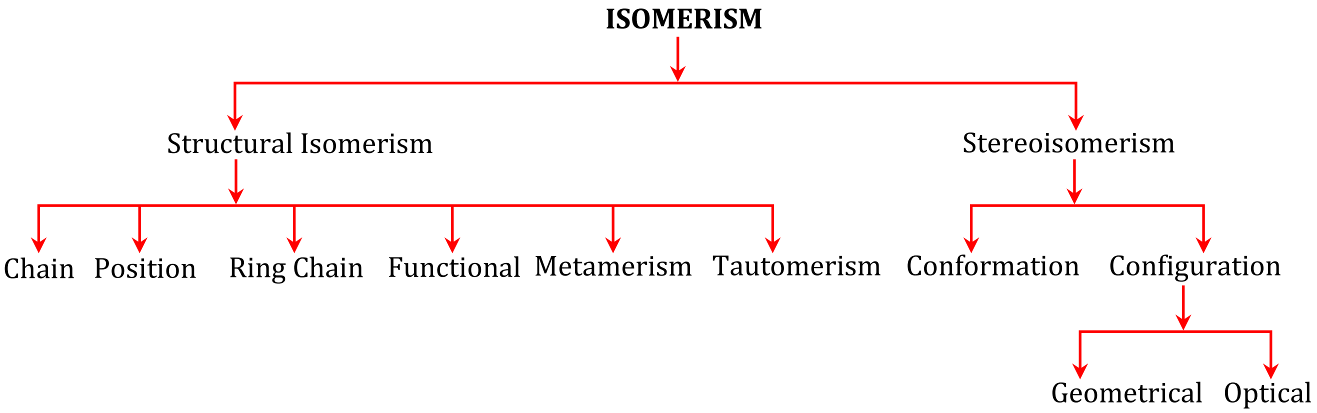
03

Isomerism

Isomerism name was given by Berzilius. Two or more than two organic compounds having the same molecular formula and molecular weight but different physical and chemical properties are called isomers and the phenomenon is called isomerism.


ISOMERISM
Structural Isomerism
Stereoisomerism
Chain
Position
Ring Chain
Functional
Metamerism
Tautomerism
Conformation
Configuration
Geometrical
Optical

STRUCTURAL ISOMERISM

Structural Isomerism:

When two or more number of organic compounds have same molecular formula but different structural formula these are called structural isomers.

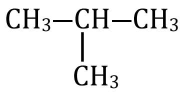
S. No.	Isomers	Characteristics	Conditions
1	Chain Isomers	They have different size of main chain or side chain	They have same functional group
2	Positional Isomers	They have different position of locant	They should have same carbon skeleton and same functional group
3	Ring Chain Isomers	They have difference in open & close chain structure	Different mode of linkage (open chain or closed chain)
4	Functional Isomers	Different functional group	Chain and positional isomerism is not considered
5	Metamerism	Different nature of alkyl group along a polyvalent functional group on either side	They should have same nature of functional group; chain & positional isomerism is ignored
6	Tautomerism	Different position of an atom usually hydrogen atom	The two isomers remains in dynamics equilibrium with each other

1. Chain Isomerism (CI) :

The compounds which have same molecular formula, same functional group, same position of functional group or multiple bond or substituent but different arrangement of carbon chain (main chain or side chain) shows chain isomerism.

Example : $CH_3-CH_2-CH_2-CH_3$

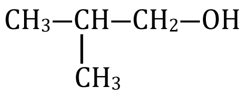
Butane (4C)


$CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_3$

2-Methyl propane(3C)

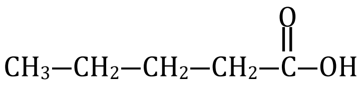
Example : $CH_3-CH_2-CH_2-CH_2-OH$

1-Butanol (4C)

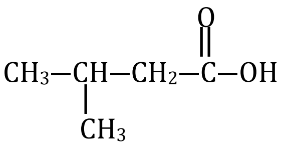

$CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-OH$

2-Methyl-1-propanol(3C)

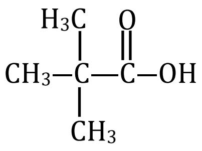
Example :


$CH_3-CH_2-CH_2-CH_2-\overset{\overset{O}{||}}{C}-OH$

Pentanoic acid (5C)


$CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-\overset{\overset{O}{||}}{C}-OH$

2-Methyl butanoic acid


$CH_3-\overset{\overset{H_3C}{|}}{\underset{\underset{CH_3}{|}}{C}}-\overset{\overset{O}{||}}{C}-OH$

2,2-Dimethyl propanoic acid

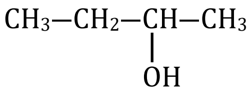
2. Position Isomerism (PI) :

The compounds which have same molecular formula, same functional group, same parent carbon chain but different position of functional group or multiple bond or substituents, are position isomers.

Example : $CH_2=CH-CH_2-CH_3$ $CH_3-CH=CH-CH_3$

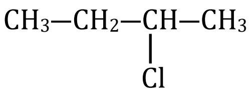
But–1–ene But–2–ene

Example : $CH_3-CH_2-CH_2-CH_2-OH$


$CH_3-CH_2-\underset{\underset{OH}{|}}{CH}-CH_3$

1–Butanol 2–Butanol

Example : $CH_3-CH_2-CH_2-CH_2-Cl$


$CH_3-CH_2-\underset{\underset{Cl}{|}}{CH}-CH_3$

1–Chlorobutane 2–Chlorobutane

Illustration 1

How many structural isomers are possible for molecular formula C_6H_{14}

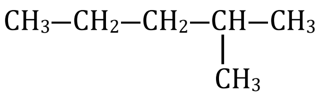
Solution:

C_6H_{14} has 5 isomers :

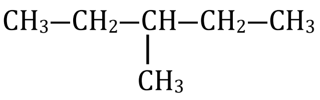
(a) $CH_3CH_2CH_2CH_2CH_2CH_3$

Hexane

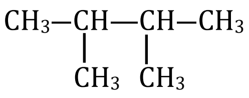
(b)


$CH_3-CH_2-CH_2-\underset{\underset{CH_3}{|}}{CH}-CH_3$

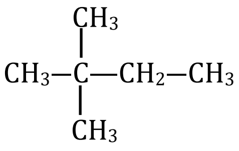
2–Methyl pentane

(c) 
$CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_3$

3–Methyl pentane

(d) 
$CH_3-\underset{\underset{CH_3}{|}}{CH}-\underset{\underset{CH_3}{|}}{CH}-CH_3$

2,3–Dimethyl butane

(e) 
$CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_2-CH_3$

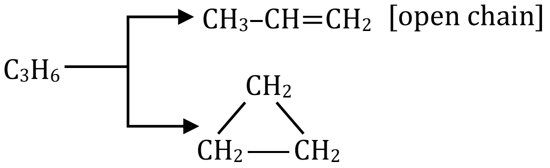
2,2–Dimethyl butane

a & b, b & d, a & c, c & d ⟶ Chain Isomers

b & c, d & e ⟶ Position Isomers

3. Ring chain isomerism (RCI) :

Same molecular formula but different mode of linking (open chain or closed chain) of carbon atoms.



[closed chain or ring]

They have same molecular formula so they are Ring chain isomers.

Illustration 2

How many minimum carbons required for Chain isomerism and Position isomerism in alkenes?

Solution:

4, 4

Illustration 3

How many minimum carbons required for Chain isomerism and Position isomerism in alkynes?

Solution:

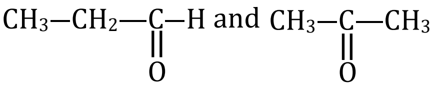
5, 4

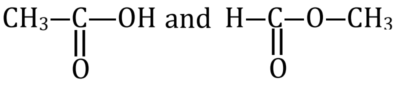
4. Functional Isomerism :-

Same molecular formula but different functional groups.

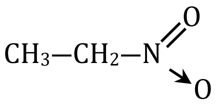
Following compounds show Functional isomerism, as they have same molecular formula and different functional group.

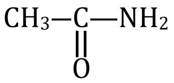
(i) Alcohol and ether ⟶ CH_3-CH_2-OH and CH_3-O-CH_3

(ii) Aldehydes and ketones ⟶ 
$CH_3-CH_2-\underset{\underset{O}{||}}{C}-H$ and $CH_3-\underset{\underset{O}{||}}{C}-CH_3$

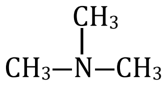
(iii) Acids and ester ⟶ 
$CH_3-\underset{\underset{O}{||}}{C}-OH$ and $H-\underset{\underset{O}{||}}{C}-O-CH_3$

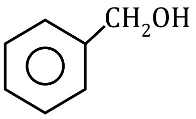
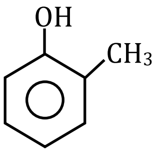
(iv) Cyanide and isocyanide ⟶ $CH_3-CH_2-CH_2-CN$ and $CH_3-CH_2-CH_2-NC$

(v) Nitro and Nitrite ⟶ 
$CH_3-CH_2-N\begin{matrix}/\!\!/O\\ \searrow O\end{matrix}$ and CH_3-CH_2-O-NO

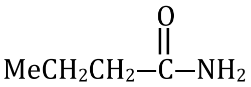
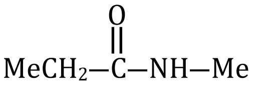
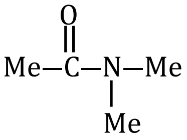
(vi) Amide and Oxime ⟶ 
$CH_3-\underset{\underset{O}{\|}}{C}-NH_2$ and $CH_3-CHNOH$ and $CH_3CH_2-N=O$

(vii) 1°, 2°, 3° amines

(i) $CH_3-CH_2-CH_2-NH_2$ (ii) $CH_3-NH-CH_2-CH_3$ (iii) 
$CH_3-\overset{\overset{CH_3}{|}}{N}-CH_3$

(viii) Alcoholic and phenolic compounds :  and 

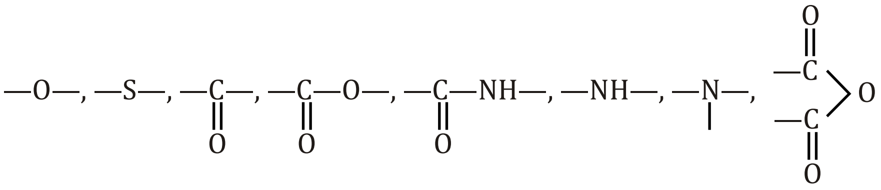
(ix) 1°, 2°, 3° amide


$MeCH_2CH_2-\overset{\overset{O}{\|}}{C}-NH_2$,

$MeCH_2-\overset{\overset{O}{\|}}{C}-NH-Me$,

$Me-\overset{\overset{O}{\|}}{C}-\underset{\underset{Me}{|}}{N}-Me$

5. Metamerism:

Same molecular formula, same polyvalent Functional group but different alkyl groups attached to polyvalent Functional group on either side.

Polyvalent Functional group [More than one valency] are :


$-O-,\ -S-,\ -\underset{\underset{O}{\|}}{C}-,\ -\underset{\underset{O}{\|}}{C}-O-,\ -\underset{\underset{O}{\|}}{C}-NH-,\ -NH-,\ -\underset{|}{N}-,\ \begin{matrix}-\overset{\overset{O}{\|}}{C}\\ -\underset{\underset{O}{\|}}{C}\end{matrix}\!\!>O$

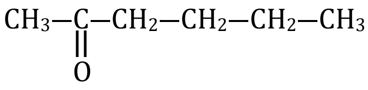
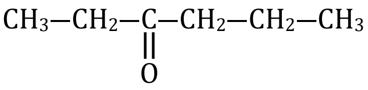
Example : $CH_3-O-CH_2-CH_2-CH_3$; $CH_3-CH_2-O-CH_2-CH_3$

Both are metamers

Example : $CH_3-CH_2-NH-CH_2-CH_3$ $CH_3-NH-CH_2-CH_2-CH_3$

N–Ethyl ethanamine N–Methyl propanamine

They are only metamers not CI

Example : 
$CH_3-\underset{\underset{O}{\|}}{C}-CH_2-CH_2-CH_2-CH_3$

$CH_3-CH_2-\underset{\underset{O}{\|}}{C}-CH_2-CH_2-CH_3$

2–Hexanone 3–Hexanone

Both are metamers and position isomer

KEY POINTS

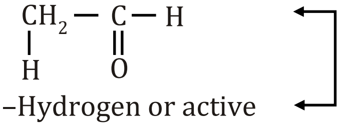
1. Parent carbon chain and functional group must be same in chain isomers.
2. Polyvalent functional group must be present for metamerism.
3. 1°, 2° and 3° amines are considered as different type of functional group.
4. 1°, 2° and 3° alcohols are considered as same functional group.

6. Tautomerism or Desmotropism:

Tautomerism was introduced by "Laar". It's also called desmotropism.

Defination of tautomerism → Arises due to rapid oscillation of an atom usually hydrogen between 2 polyvalent atoms in a molecule.

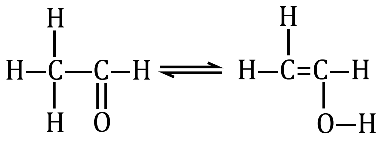
The tautomerism is also called **kryptomerism or allotropism or desmotropism or dynamic isomerism.**



$\underset{H}{CH_2}-\underset{O}{\underset{\|}{C}}-H$

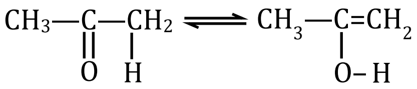
α–H attached to carbonyl compound is active H

Example :



$H-\underset{H}{\overset{H}{C}}-\underset{O}{\underset{\|}{C}}-H \rightleftharpoons H-\overset{H}{C}=\underset{O-H}{C}-H$

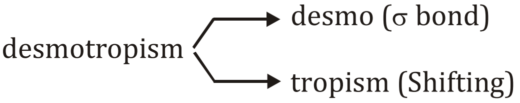
Example :



$CH_3-\underset{O}{\underset{\|}{C}}-\underset{H}{CH_2} \rightleftharpoons CH_3-\underset{O-H}{C}=CH_2$

keto ene + ol = enol

→ It involves σ–bond shifting



desmotropism → desmo (σ bond)

desmotropism → tropism (Shifting)

→ Tautomers remain in dynamic equilibrium with each other

→ More stable isomer is present in higher concentration.

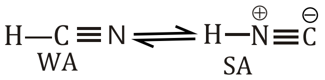
→ Tautomerism Involve intramolecular acid-base reaction.

→ The process can be catalysed by acid as well as base

Type of tautomerism:

(1) Diad tautomerism :-

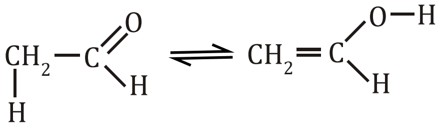
If the migration take place in between adjacent atoms then system is known as diad tautomerism.



$\underset{WA}{H-C\equiv N} \rightleftharpoons \underset{SA}{H-\overset{\oplus}{N}\equiv\overset{\ominus}{C}}$

(2) Triad tautomerism :-

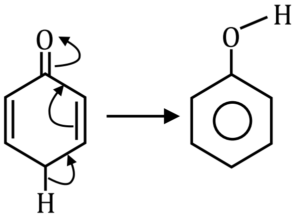
If migration take place between first atom to third atom.



Ket enol

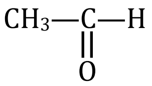
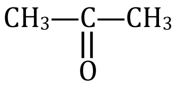
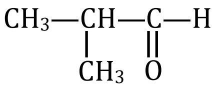
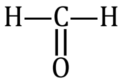
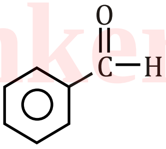
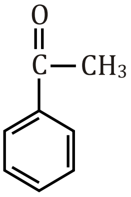
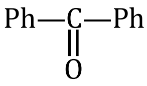
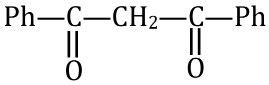
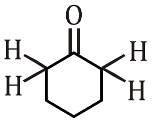
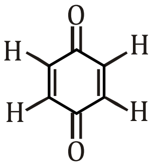
(3) Space tautomerism :

If distance of migration is more than 3 atom then it is known as space tautomerism.

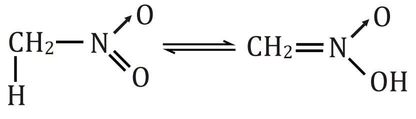


Condition for Tautomerism :

(a) For carbonyl compounds :- Carbonyl compounds having atleast one α–H show tautomerism

(i)	 $CH_3-\underset{\overset{\parallel}{O}}{C}-H$	3 α H	show tautomerism.
(ii)	 $CH_3-\underset{\overset{\parallel}{O}}{C}-CH_3$	6 α H	show tautomerism
(iii)	 $CH_3-\underset{\overset{\mid}{CH_3}}{CH}-\underset{\overset{\parallel}{O}}{C}-H$	1 α H	show tautomerism
(iv)	 $H-\underset{\overset{\parallel}{O}}{C}-H$	No α H	No tautomerism
(v)		No α H	No Tautomerism
(vi)		3 α H	shows tautomerism (Acetophenone)
(vii)	 $Ph-\underset{\overset{\parallel}{O}}{C}-Ph$	No α H	No tautomerism (Benzophenone)
(viii)	 $Ph-\underset{\overset{\parallel}{O}}{C}-CH_2-\underset{\overset{\parallel}{O}}{C}-Ph$	2 α H	shows tautomerism
(ix)		4 α H	shows tautomerism
(x)		4 α H	attached sp^2 carbon does not take part in tautomerism

(b) For nitro compounds: Nitro compounds having atleast one α – H show tautomerism

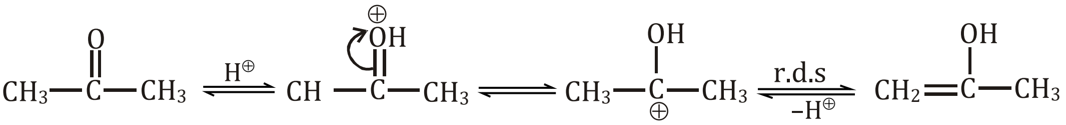


$$\underset{H}{\underset{|}{CH_2}}-N\overset{\nearrow O}{\searrow\!\!\!\!= O} \rightleftharpoons CH_2=N\overset{\nearrow O}{\searrow OH}$$

Nitro form Aci nitro form

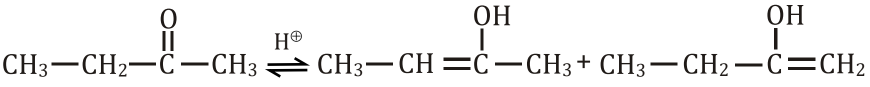
Mechanism:

(I) Acid catalysed Mechanism:



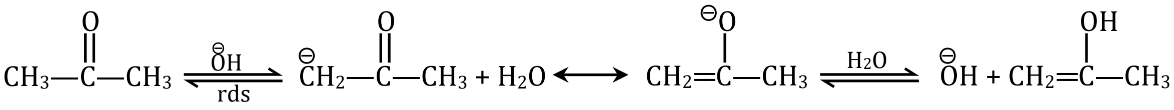
$$CH_3-\overset{O}{\overset{\|}{C}}-CH_3 \xrightleftharpoons{H^{\oplus}} CH-\overset{\overset{\oplus}{OH}}{\overset{\|}{C}}-CH_3 \rightleftharpoons CH_3-\underset{\oplus}{\overset{OH}{\overset{|}{C}}}-CH_3 \xrightleftharpoons[-H^{\oplus}]{r.d.s} CH_2=\overset{OH}{\overset{|}{C}}-CH_3$$

→ More stable product is major product in case of more than one enol

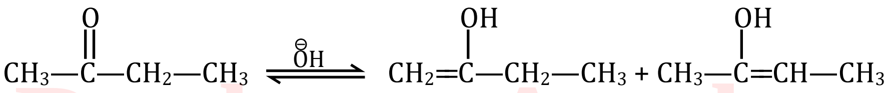


$$CH_3-CH_2-\overset{O}{\overset{\|}{C}}-CH_3 \xrightleftharpoons{H^{\oplus}} CH_3-CH=\overset{OH}{\overset{|}{C}}-CH_3+CH_3-CH_2-\overset{OH}{\overset{|}{C}}=CH_2$$

(II) Base catalysed mechanism :



$$CH_3-\overset{O}{\overset{\|}{C}}-CH_3 \xrightleftharpoons[rds]{\overset{\ominus}{O}H} \overset{\ominus}{C}H_2-\overset{O}{\overset{\|}{C}}-CH_3+H_2O \longleftrightarrow CH_2=\overset{\ominus O}{\overset{|}{C}}-CH_3 \xrightleftharpoons{H_2O} \overset{\ominus}{O}H+CH_2=\overset{OH}{\overset{|}{C}}-CH_3$$



$$CH_3-\overset{O}{\overset{\|}{C}}-CH_2-CH_3 \xrightleftharpoons{\overset{\ominus}{O}H} CH_2=\overset{OH}{\overset{|}{C}}-CH_2-CH_3+CH_3-\overset{OH}{\overset{|}{C}}=CH-CH_3$$

Enol Content : Generally keto form is more stable than enol form

→ % amount of enol at equilibrium is known as enol content.

→ enol content ∝ stability of enol

∝ aromatization of enol

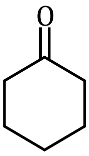
∝ stability of enolate ion

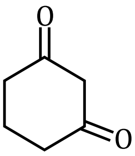
∝ Intramolecular H-bonding

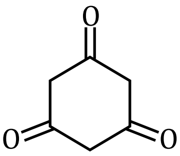
∝ Stability of anion

Illustration 4

Arrange following in decreasing order of enol content.

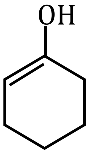


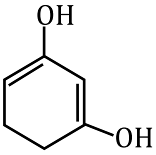




Ans. |||>||>|

Solution:





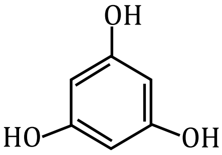
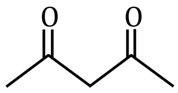
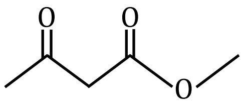
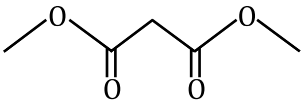


Illustration 5

Arrange following in decreasing order of enol content.



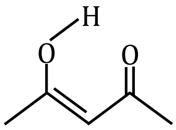


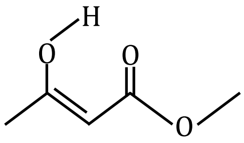


Ans. | > || > |||

Solution:

Due to strong chelation in I





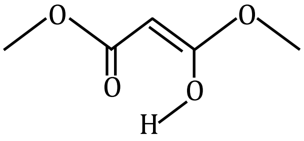
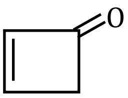
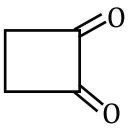
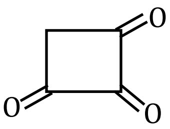


Illustration 6

Arrange following in decreasing order of enol content.







Ans.

Aromatic

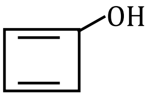
↑

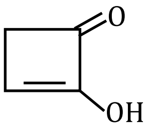
III > II > I → Anti aromatic

↓

Non aromatic

Solution:





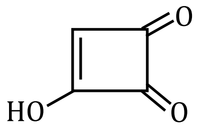
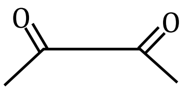
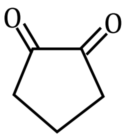


Illustration 7

Arrange following in decreasing order of enol content.

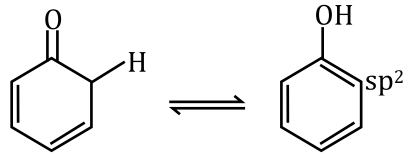




Ans. |<||

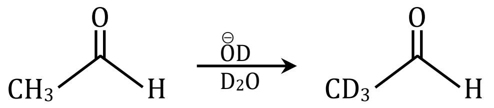
Illustration 8

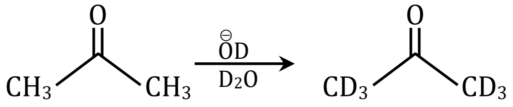
Arrange following in decreasing order of enol content.

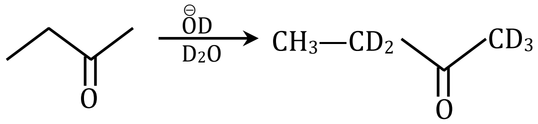


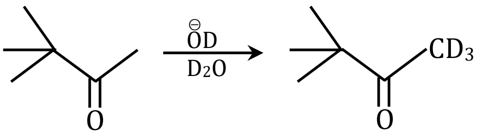
"keto" (≈1%) "enol" (stable by resonance and aromatic nature) (≈99%)

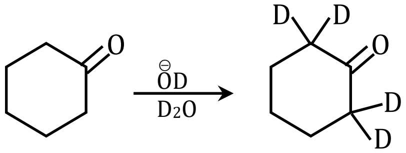
Kinetic isotopic exchange:

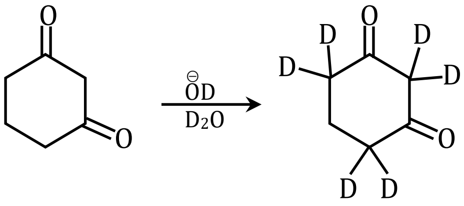
1.


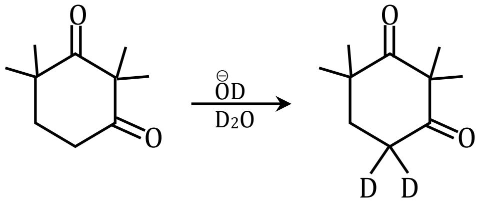
2.


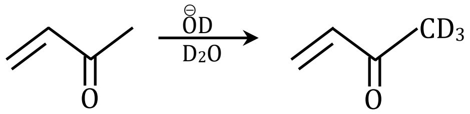
3.


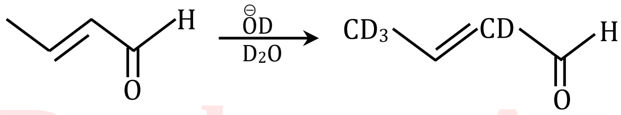
4.


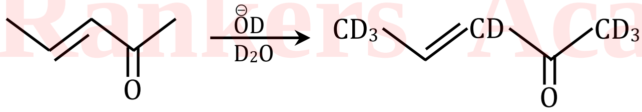
5.


6.


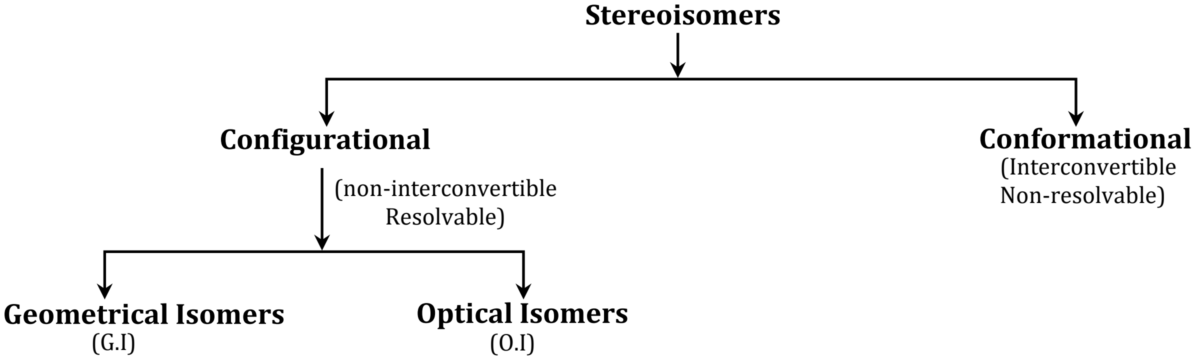
7.


8.


9.


10.


Stereoisomerism:


Stereoisomers
Configurational
(non-interconvertible
Resolvable)
Conformational
(Interconvertible
Non-resolvable)
Geometrical Isomers
(G.I)
Optical Isomers
(O.I)

Stereoisomerism :

Two or more than two compounds having same molecular formula, same structural formula but different arrangements of atoms or groups in space.

GEOMETRICAL ISOMERISM AND CONFORMATIONS

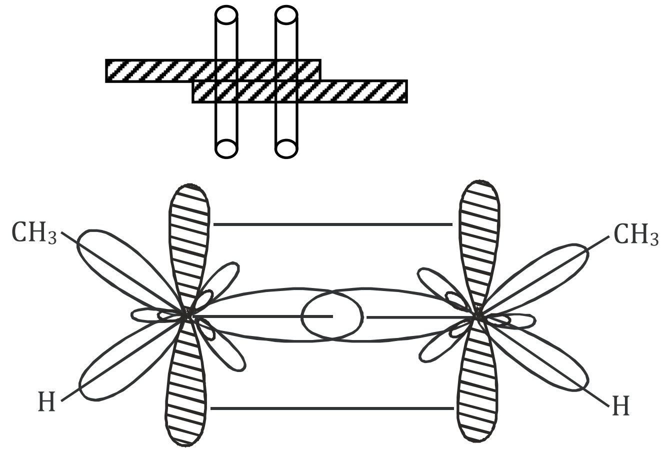
Geometrical isomerism:

Stereo-isomers which are obtained due to different orientation of atoms or group in the space around restricted rotation are known as G.I.

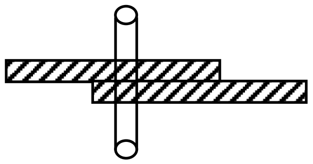
Condition for Geometrical isomerism:

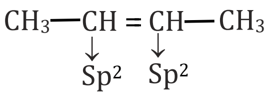
Condition 1 : Restricted rotation must be present.

Restricted Rotation



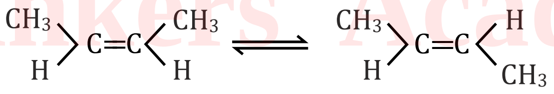
Free Rotation





$CH_3—\underset{Sp^2}{\underset{\downarrow}{CH}} = \underset{Sp^2}{\underset{\downarrow}{CH}}—CH_3$

Example :



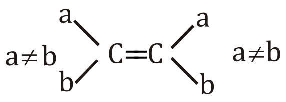
cis–2–butene trans–2–butene

Note :

cis $\rightleftharpoons$ trans is possible only when π bond break.

Condition 2

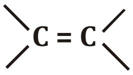
(i) The two groups at each end of restricted bond must be different.

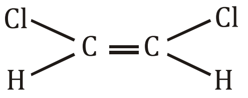
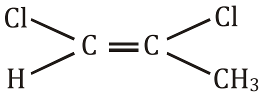
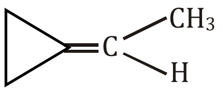
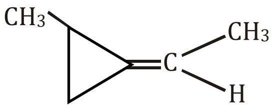


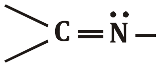
Example	**G.I.**
$C_{aa}=C_{aa}$	×
$C_{aa}=C_{bd}$	×
$C_{aa}=C_{bb}$	×
$C_{ab}=C_{ab}$	✓
$C_{ab}=C_{bd}$	✓
$C_{ab}=C_{de}$	✓

Condition 3

Terminal valency should be present in the same plane.

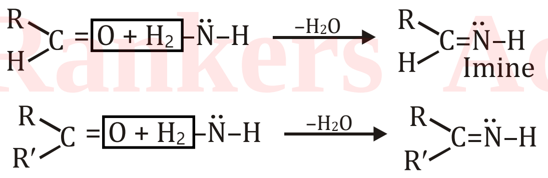
1. **Geometrical isomerism in alkene** 

Example	G.I.
	Yes
	Yes
	No
	Yes

2. **Geometrical isomerism in** 

(a) Imine ($>C=\ddot{N}-H$)

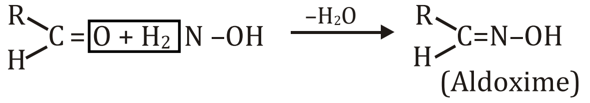
Imine compounds are produced from carbonyl compounds on reaction with ammonia.

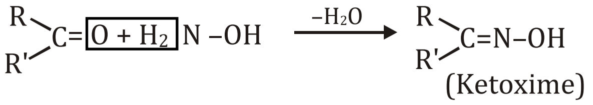

R
H
C = O + H₂ N̈–H
–H₂O
R
H
C=N̈–H
Imine
R
R′
C = O + H₂ N̈–H
–H₂O
R
R′
C=N̈–H

Imines prepared from unsymmetrical aldehydes and ketones, always show geometrical isomerism.

(b) Oximes ($>C=N-OH$)

They are prepared by reacting carbonyl compound with hydroxyl amine (NH_2-OH)

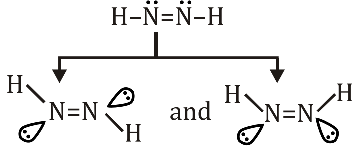

R
H
C = O + H₂ N –OH
–H₂O
R
H
C=N–OH
(Aldoxime)


R
R'
C= O + H₂ N –OH
–H₂O
R
R'
C=N–OH
(Ketoxime)

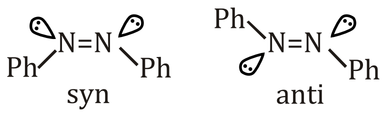
→ Unsymmetrical ketone forms two oximes.

3. **Geometrical isomerism in azo compounds (–N=N–)**

(i) H–N=N–H(H_2N_2)

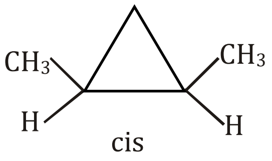

H–N̈=N̈–H
H
N=N
H
and
H
N=N
H

(ii) Ph_2N_2 (Azobenzene)

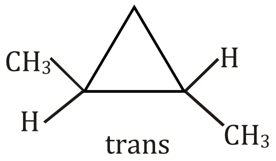

N=N
Ph
Ph
syn
Ph
N=N
Ph
anti

4. Geometrical isomerism in cyclic compound

(i)



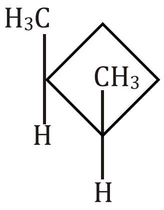
cis



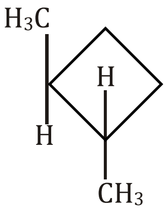
trans

Restricted rotation

(ii)

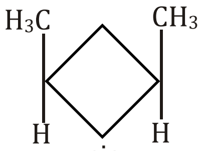


cis

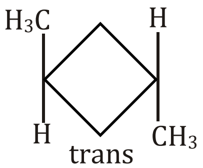


trans

(iii)



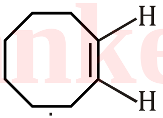
cis



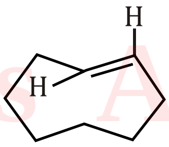
trans

5. Geometrical isomerism in cycloalkenes in double bonds

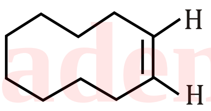
In cycloalkenes, G.I. exists across double bonds with ring size equal to or greater then 8 carbon atoms (due to ring strain)



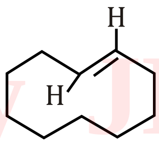
cis



trans

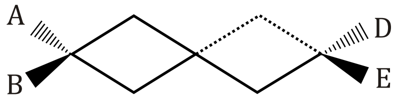
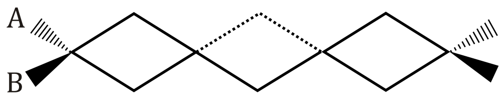


cis

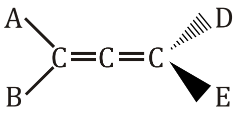
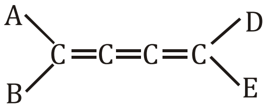


trans

6. G.I. in spiro compounds

Example	G.I.
	NO
	Yes

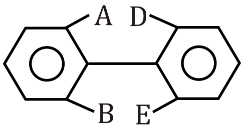
7. G.I. in cumulenes

Example	G.I.
	NO
	Yes

Note :

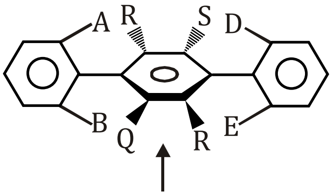
Even π-bond – don't show G.I.
Odd π-bond - show G.I.

8. G.I. in Biphenyl Compound



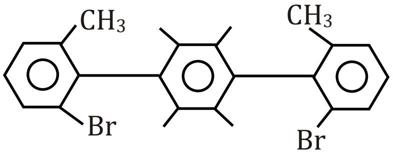
A,B,D & E : small ⇒ Conformation (Not show G.I.) ⇒ No restricted rotation
large ⇒ due to SIR (Not show G.I.) ⇒ restricted rotation
⇒ no. G.I.

→ **Triphenyl Compound**



Not in plane ⇒ as due to most stable.

Example.

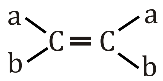

CH3
CH3
Br
Br

Show G.I.

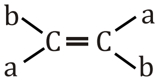
Nomenclature systems of Geometrical isomers:

(a) Cis–Trans System :

If similar groups at same side then cis and if same groups at different side then trans.

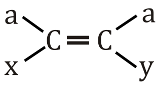


[Same groups, same side]
cis

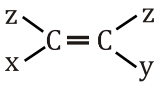


[Same groups, different side]
trans

Example :

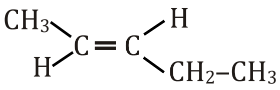


cis

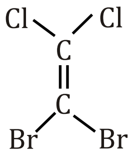


cis

Example :


CH3
H
H
CH2–CH3
C=C

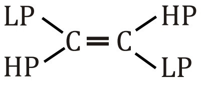
trans-2-pentene


Cl
Cl
C
C
Br
Br

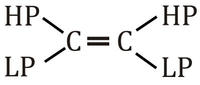
no cis–trans

(b) E – Z System :

E (Entgegen) : When high priority groups are opposite side.
Z (Zussamen) : When high priority groups are same side.


LP
HP
HP
LP
C=C

'E'


HP
HP
LP
LP
C=C

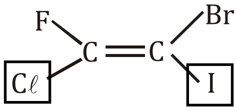
'Z'

HP – High priority and LP – Low priority

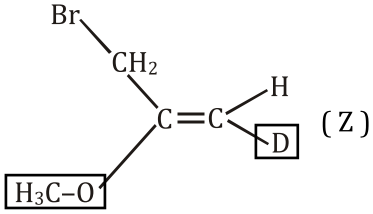
Priority Rules:

Cahn, Ingold, Prelog (CIP Rule)

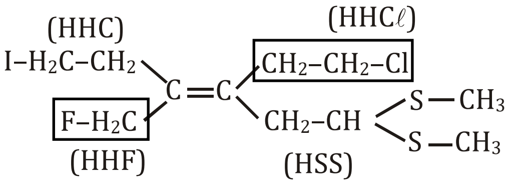
Rule 1 : Priority α Atomic no. of atom which is directly attached to restrided rotatory system.



Rule 2 : Priority α Atomic weight (in case of isotopes)

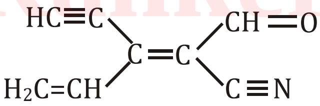

Br
CH2
H
C=C
D
(Z)
H3C–O

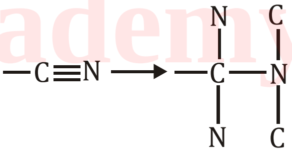
Rule 3 : If at the first directly attached, R–1 fails move to next atom and atoms attached to it.

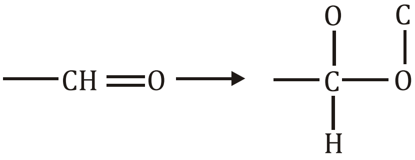

(HHC)
I–H2C–CH2
(HHCℓ)
CH2–CH2–Cl
C=C
F–H2C
(HHF)
CH2–CH
S–CH3
S–CH3
(HSS)

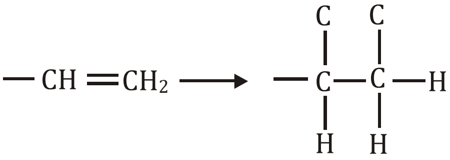
Rule 4 : If directly attached groups contain double bond then duplicate them by dummy atoms.

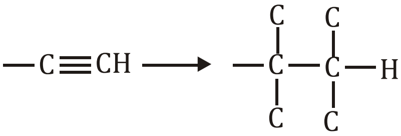
Example :


HC≡C
CH=O
C=C
H2C=CH
C≡N

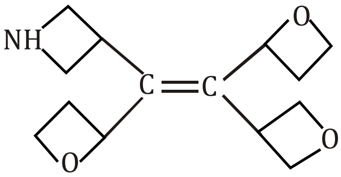


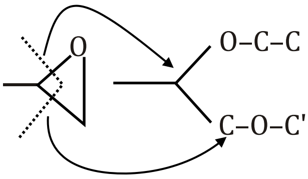


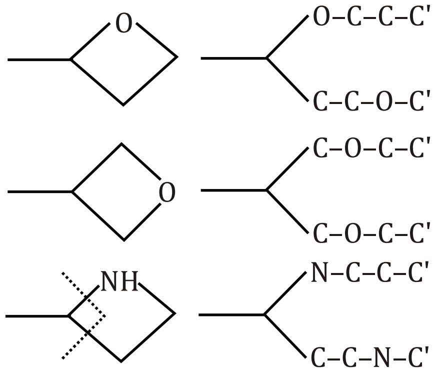

–CH=CH2
C
C
C
C
H
H
H


–C≡CH
C
C
C
C
H
C
C

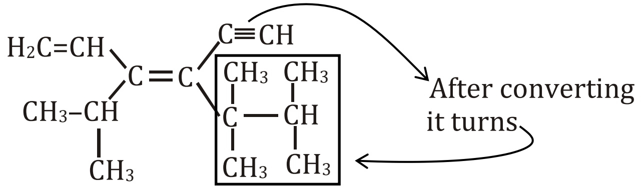
Example :



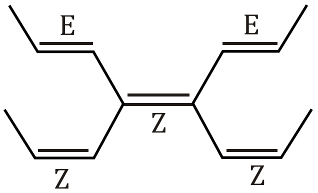



O
O–C–C–C'
C–C–O–C'
O
C–O–C–C'
C–O–C–C'
NH
N–C–C–C'
C–C–N–C'

Rule 5 : If after converting a multiple bond, it converts identical to original, higher priority to original one.


H₂C=CH
C=C
C≡CH
CH₃–CH
CH₃
CH₃ CH₃
C — CH
CH₃ CH₃
After converting
it turns

Rule 6 :

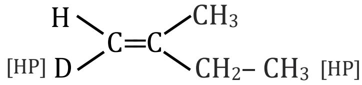


(Z)

cis/trans ⇒ cis
Z/E ⇒ Z
R/S ⇒ R

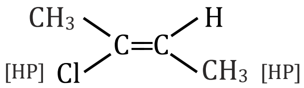
Example : Assign E-Z nomenclature

(i)


H
C=C
CH₃
[HP] D
CH₂– CH₃ [HP]

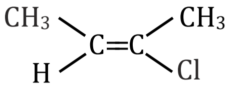
'Z'

(ii)


CH₃
C=C
H
[HP] Cl
CH₃ [HP]

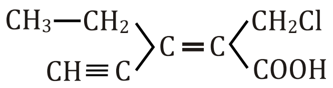
'Z'

(iii)


CH₃
C=C
CH₃
H
Cl

'E'

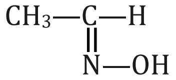
(iv)


CH₃—CH₂
C=C
CH₂Cl
CH≡C
COOH

[Cl, H, H] [HP]
[O, O, O] [LP] ; 'E'

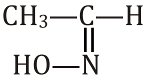
Syn-Anti Nomenclature:

Example : Acetaldoximes has two Geometrical isomers –



$$CH_3—\underset{\underset{N—OH}{||}}{C}—H$$

syn

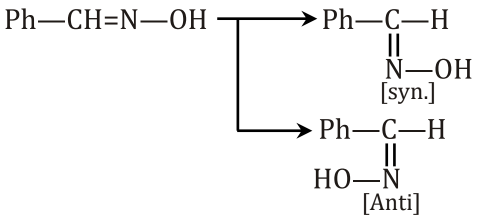


$$CH_3—\underset{\underset{HO—N}{||}}{C}—H$$

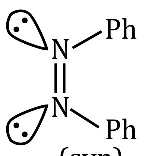
anti

When H and OH are on the same side When H and OH are on the opposite side

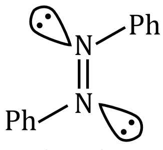
Example :



Benzaldoxime



(syn)



(Anti)

Ph—N=N—Ph (Azo benzene)

Number of Geometrical isomers: (For compounds not showing optical isomerism)

Calculation of GI :

n = no. of unit which can show GI

Case-I : Compound is unsymmetrical=2^n

Case-II : Compound is symmetrical =$2^{n-1} + 2^{p-1}$ (p is $\frac{n}{2}$ if n is even and $\frac{n+1}{2}$ if n is odd)

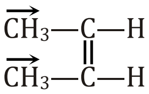
Physical properties of Geometrical isomers:

Physical Properties of Cis–Trans Isomers :

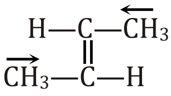
S.No.	Physical properties	Comparison	Remarks
1	Dipole moment	cis > trans	cis-isomer has resultant of dipoles while in trans isomer dipole moments cancel out
2	Boiling point	cis > trans	Molecules having higher dipole moment have higher boiling point due to larger intermolecular force of attraction
3	Solubility (in H_2O)	cis > trans	More polar molecules are more soluble in H_2O
4	Melting point	trans > cis	More symmetric isomers have higher melting points due to better packing in crystalline lattice & trans isomers are more symmetric than cis.
5	Stability	trans > cis	The molecule having more Vander Waal strain are less stable. In cis isomer the bulky group are closer they have larger Vander Waal strain.

Dipole moment [μ] :

Example :

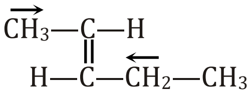


cis μ≠0

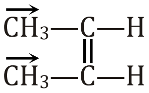


trans μ=Zero

Example :

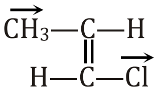


trans μ≠0



cis μ≠0

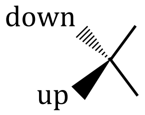
Example :



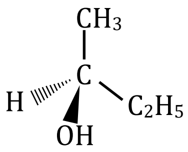
trans μ≠0

Projection Formula of Chiral Molecules:

(i) Wedge-Dash Projection formula



Example : Butan-2-ol



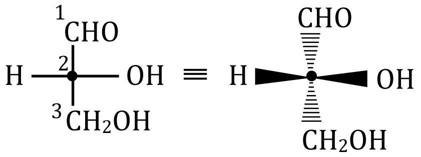
(ii) Ficher Projection formula

Rules of writing Fisher Projection formula

(a) Groups at Vertical line are away from observer.

(b) Groups at Horizontal line are towards the observer.

(c) High priority group lies at the top of vertical line (Numbering starts from top).



KEY POINTS

1. Cycloheptene and lower cycloalkenes cannot exist in trans form. So they cannot show G.I.
2. Allenes which have odd number of double bonds can show G.I.
3. Double bond is not necessary condition for G.I. Restricted rotation is necessary condition for G.I.

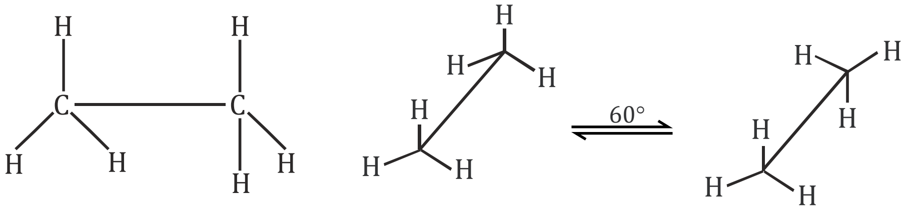
Conformational Isomerism

The different arrangement of atoms in space that result from the free rotation around C—C bond axis are called conformations. The phenomenon is called conformational isomerism

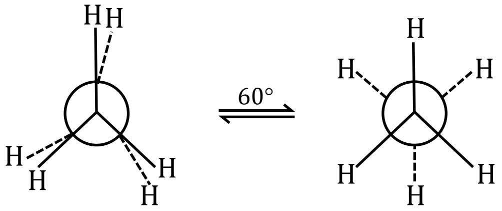
Some Important Terms :

- **(i) Dihedral or Torsional angle :** Angle between bonds of two adjacent atoms is knows as dihedral or Torsional angle.
- **(ii) Staggered Conformation:** Conformations in which valency triangle are opposite and most priority group are also opposite then known as staggered/anti conformation.
- **(iii) Eclipsed Conformation**: In which two valency triangle as well as most priority group (according to CIP) are superimpose then known as Eclipsed conformation.
- **(iv) Skew Conformation :** All other conformations are known as Skew Conformations.
- **(v) Torsional Strain :** Is the name given to the repulsion felt by the bonded electron of front carbon atom with the bonded electrons of back carbon atom.
- **(vi) Steric Strain :-** Strain felt by the repulsion of atoms or groups of atoms of front carbon atom with back carbon atom. In most circumstances torsional strain dominates over steric strain.

Conformers of ethane [CH_3—CH_3]



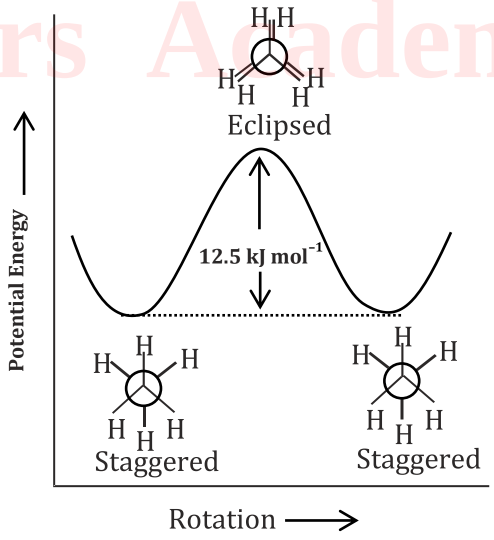
(I) (Saw horse projection) (II)



(III) (IV)

(Newman projection)

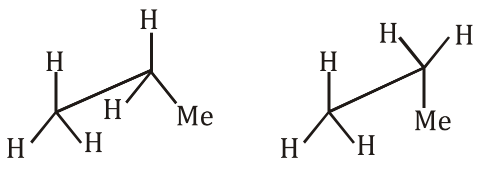
I = III (Eclipsed form) in this form distance between 2C–H bonds is minimum so maximum repulsion or minimum stable.
II = IV (Staggered form) in this form distance between 2C–H bonds is maximum so minimum repulsion so maximum stable.
There are infinite conformers between eclipsed and staggered forms which are called as skew forms
Stability order : Staggered > Skew > Eclipsed
The variation of energy with rotation about the C–C bond in ethane has been shown in figure below :



Changes in energy during rotation about C–C bond in

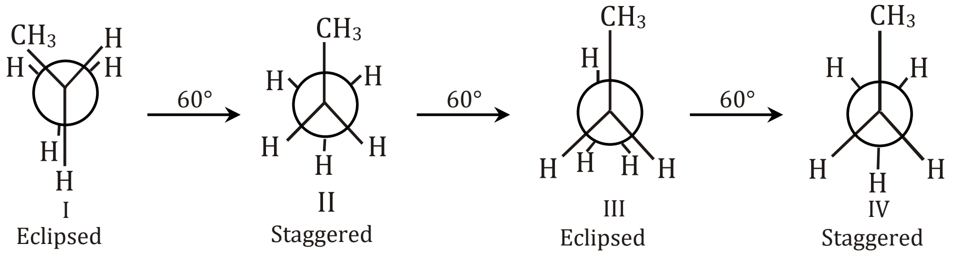
Conformation of propane :
Ex. $CH_3\rangle\!-CH_2-CH_3$

Saw horse projection Formula :

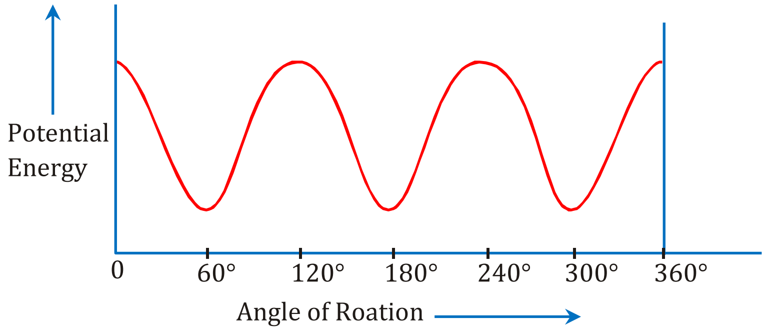


Eclipsed Staggered

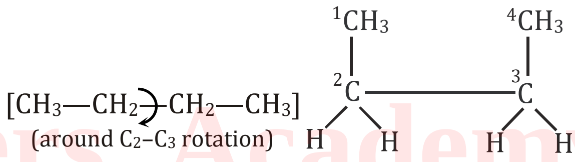
Newman Projection Formula :

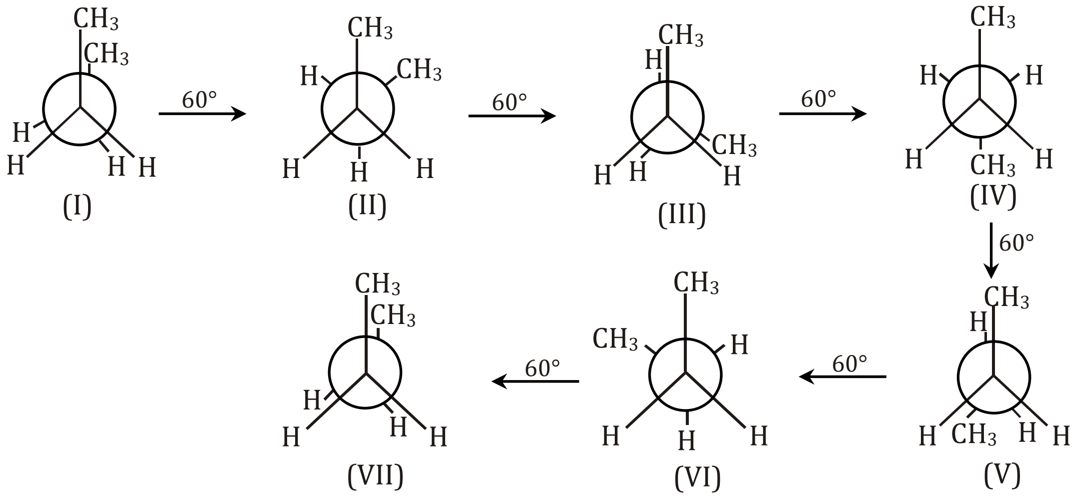

CH3
H
H
H
H
H
I
Eclipsed
60°
CH3
H
H
H
H
H
II
Staggered
60°
CH3
H
H
H
H
H
III
Eclipsed
60°
CH3
H
H
H
H
H
IV
Staggered

Propane :



Conformation of Butane :


[CH3—CH2—CH2—CH3]
(around C2–C3 rotation)
1CH3
4CH3
2C
3C
H
H
H
H


CH3
CH3
H
H
H
H
(I)
60°
CH3
H
CH3
H
H
H
(II)
60°
CH3
H
CH3
H
H
H
(III)
60°
CH3
H
H
H
CH3
H
(IV)
60°
CH3
H
H
CH3
H
H
(V)
60°
CH3
CH3
H
H
H
H
(VI)
60°
CH3
CH3
H
H
H
H
(VII)

I/VII = Fully eclipsed

II/VI = Gauche form

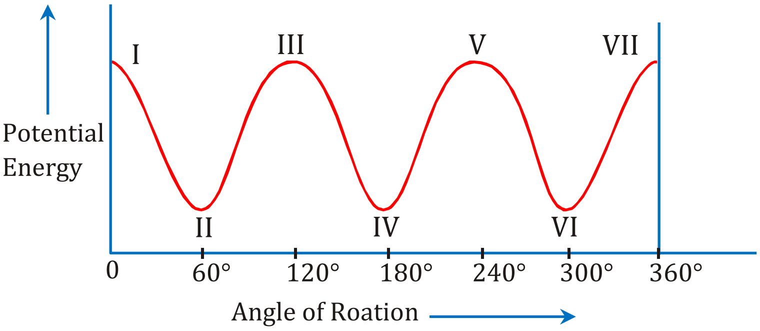
III/V = Partially eclipsed

IV = Anti form

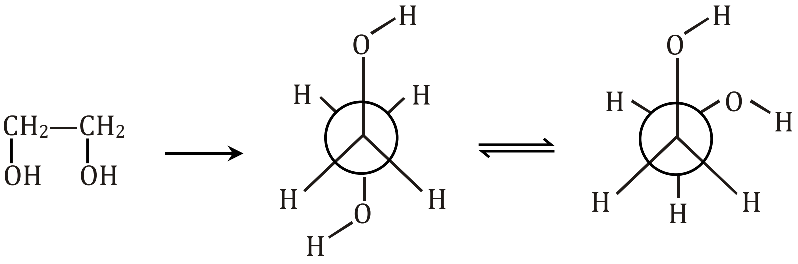
Stability Order IV > II > III > I

P.E. Order IV < II < III < I

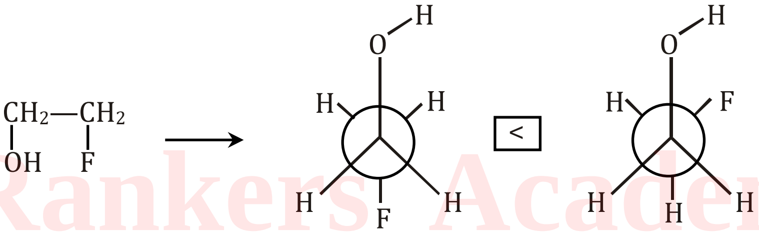
Potential Energy Diagram :

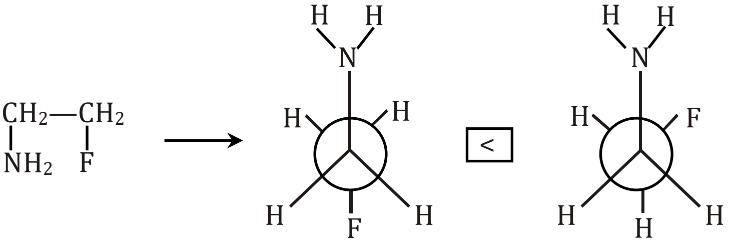


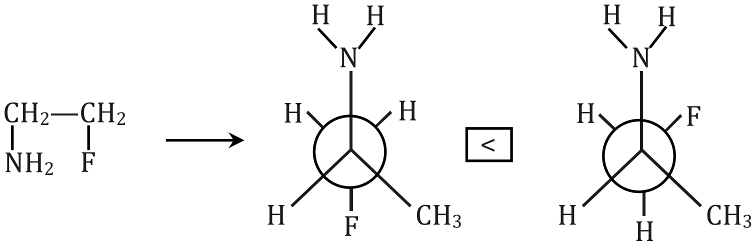
1. Draw most stable conformation of following:

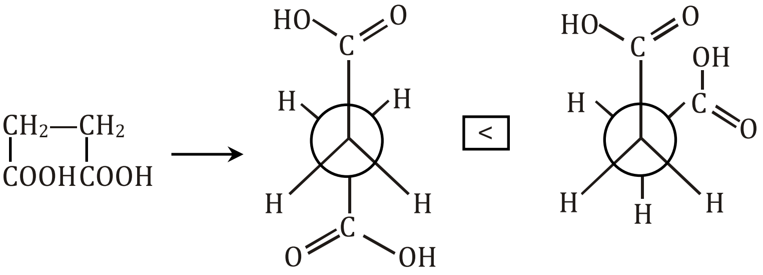
(i)

CH2—CH2
OH OH
O H
H
H
O
H

Due to H-bonding gauche is more stable than anti

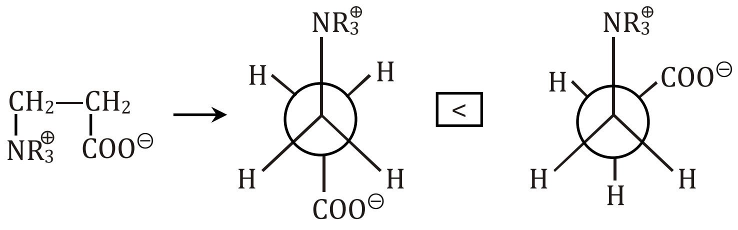
(ii)


(iii)

CH2—CH2
NH2 F
<

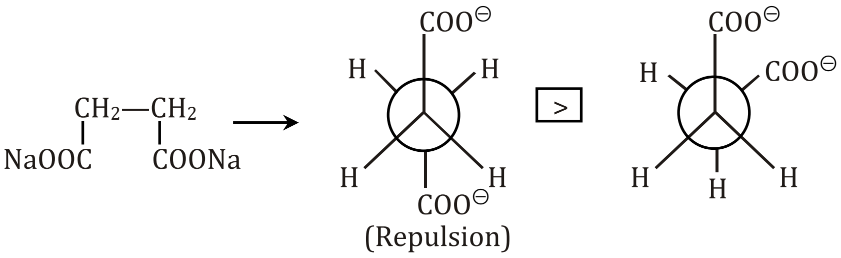
(iv)

CH2—CH2
NH2 F
CH3
<

(v)

CH2—CH2
COOH COOH
HO
C
O
OH
<

(vi)


CH2—CH2
NR3⊕ COO⊖
NR3⊕
H
H
H
H
COO⊖
<
NR3⊕
H
COO⊖
H
H
H

(vii)


CH2—CH2
NaOOC COONa
COO⊖
H
H
H
H
COO⊖
(Repulsion)
>
COO⊖
H
COO⊖
H
H
H

Dipole moment in Conformational analysis :

Dipole moment (μ) :

$\mu_{net} = \sum \mu_i x_i$

$\mu_{net} = \mu_a x_a + \mu_g x_g$

$x_a + x_g = 1$

∴ Only stable conformers are considered.

a ⇒ anti

g ⇒ gauche

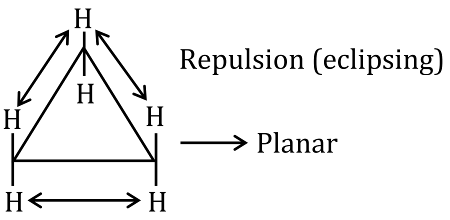
Conformational analysis of cycloalkane :

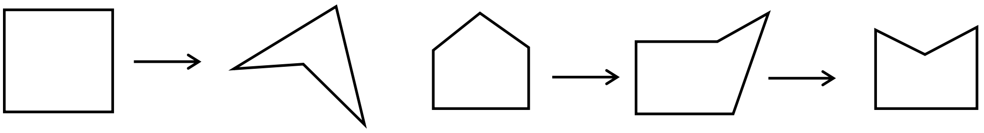
Angle strain : When acyclic compound is converted into cyclic compound then deviation in bond angle is observed and per bond deviation is known as angle strain.

$$\text{Angle strain} = \frac{\text{Total deviation}}{2}$$

When size of ring increases then it becomes flexible and partial σ bond rotation is observed so larger ring acquire non-planar orientetion to avoid eclipsing.

Eclipsing :

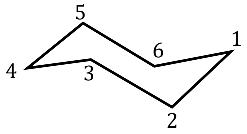

H
H
H
H
H
H
Repulsion (eclipsing)
Planar



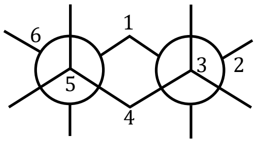
Puckered structure Envelope structure

Conformational analysis of cyclohexane :

(i) Chair form : Experimental evidences show that cyclohexane is non-planar. If we look to the models of the different conformations that are free of angle strain first there is chair conformation. If we sight along each of the carbon-carbon bonds in turn we see in every case perfectly staggered bonds.



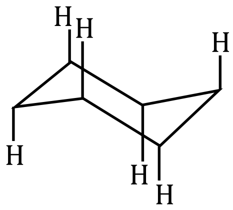
Chair conformation



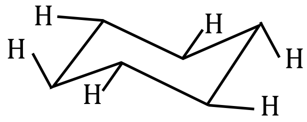
Chair conformation (all staggered bonds)

The conformation is thus free of all the strains, it lies at energy minimum and is therefore a conformational isomer. The chair form is the most stable conformation of cyclohexane.

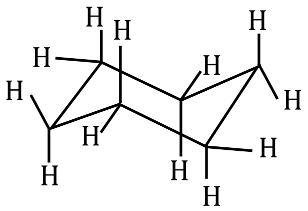
Axial and equatorial bonds in chair form of cyclohexane :



Axil C–H bonds



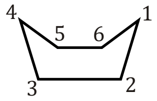
Equatorial C–H bonds



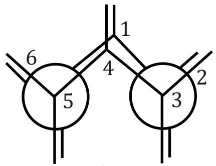
Axial & equatorial bonds together

The 12 hydrogen atoms of chair conformation of cyclohexane can be divided into two groups. Six of the hydrogens, called axial hydrogens, hence their bonds parallel to a vertical axis that passes through the rings centre. These axial bonds are directed up & down on adjacent carbons. The second set of six hydrogens called equatorial hydrogens are located approximately along the equator of the molecule.

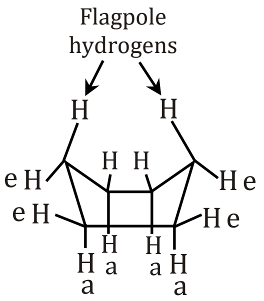
(ii) Boat form: Another conformation which is known as boat conformation has exactly eclipsed conformations.



Boat conformation

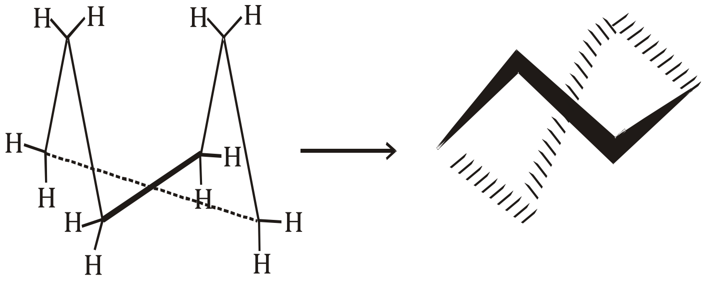


Boat conformation (all eclipsed bonds)

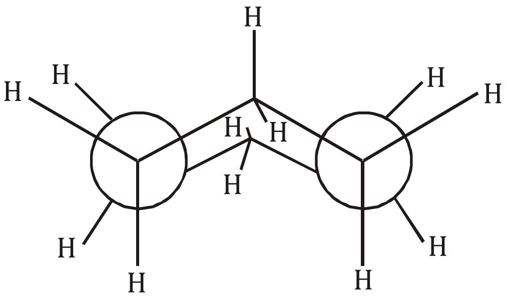


In boat form of cyclohexane 6 hydrogens are equatorial, 4 hydrogens are axial and two hydrogens are flagpoles. It is an unstable conformation of cyclohexane due to torsional strain among axial hydrogens and due to van der waals strain caused by crowding between the "flagpole" hydrogens.

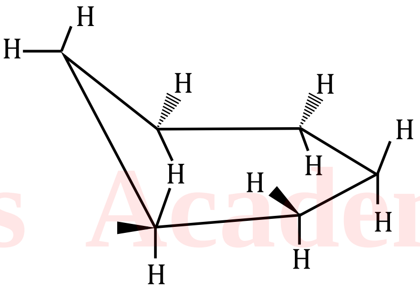
(iii) Twist Boat conformation



Newmann :

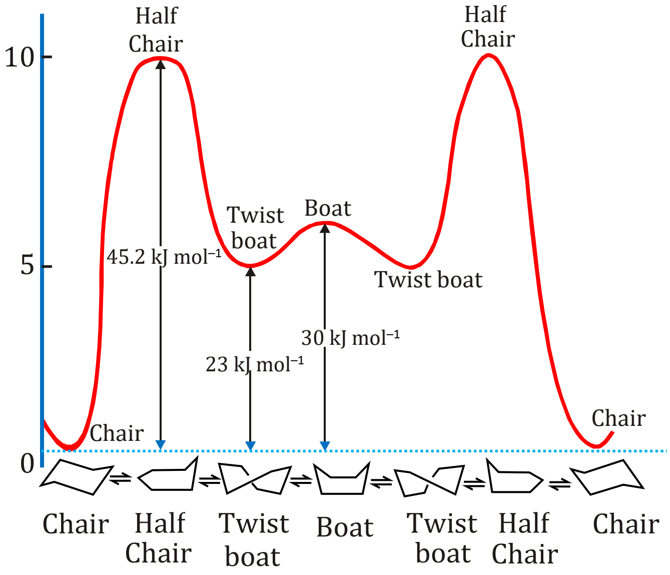


(iv) Half Chair Conformation



Stability order :

Chair > Twist boat > Boat > Half Chair

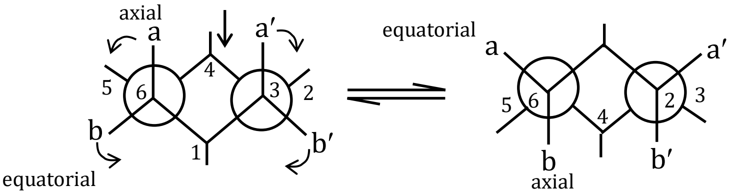


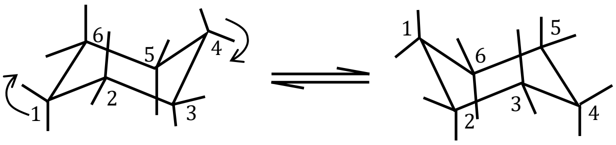
The relative energies of the various conformations of cyclohexane.

The positions of maximum energy are conformations called half-chair conformations, in which the carbon atoms of one end of the ring have become coplanar.

Conformational inversion (Ring flipping) in cyclohexane

Like alkanes cyclohexane too is conformationally mobile. Through a process known as ring inversion, Chair-chair interconversion, or more simple ring flipping one chair conformation is converted to another chair.

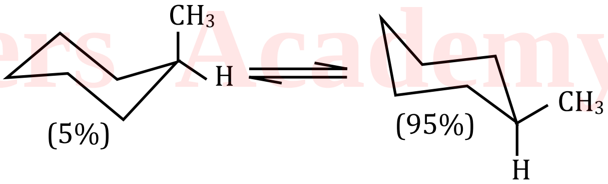




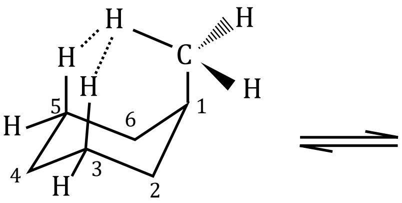
By ring flipping all axial bonds convert to equatorial and vice-versa. The activation energy for cyclohexane ring inversion is 45 kJ/mol. It is a very rapid process with a half-life of about 10^{-5} sec at 25°C.

Conformational analysis of monosubstituted cyclohexanes

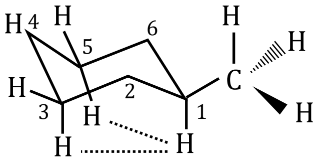
In ring inversion in methylcyclohexane the two chair conformations are not equivalent. In one chair the methyl group is axial ; in the other it is equatorial. At room temperature 95% of the methylcyclohexane exist in equatorial methyl group whereas only 5% of the molecule have an axial methyl group.



1,3-diaxial repulsion : A methyl group is less crowded when it is equatorial than when it is axial. The distance between the axial methyl groups at C-1 and two hydrogens at C-3 and C-5 is less than the sum of their vander waal radii which causes vander waal strain in the axial conformation this type of crowding is called 1,3-diaxial repulsions. When the methyl group is equatorial, it experience no significant crowding.



Vander Waals strain between hydrogen of axial CH_3 and axial hydrogens at C-3 and C-5



No Vander Waals strain between hydrogen at C-1 and axial hydrogens at C-3and C-5

OPTICAL ISOMERISM

Definitions of Optical Isomerism and Polarimetry Experiment, Optical Activity, Specific Rotation, d(+), l(-):

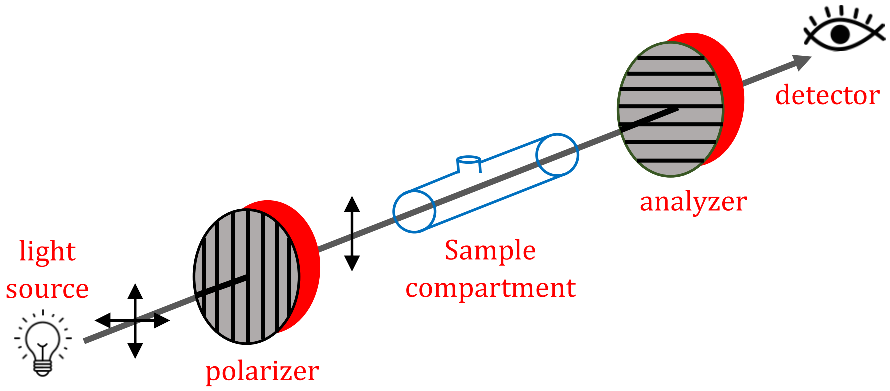
Optical Isomerism :

Compounds which have same molecular and structural formula but have different optical activity are known as optical isomers.

Optical activity :

Certain substances possess the property to rotate the plane of polarized light. Such substances are called optically active substances and this phenomenon is called optical activity.

Light has vibrations occur in all planes at right angles to the line of propagation. In plane polarized light the vibrations take place only in one plane. Plane polarized light can be obtained by passing ordinary light through a Nicol prism.



Certain organic compounds, when their solutions are placed in the path of a plane polarized light, have the remarkable property of rotating its plane through a certain angle which may be either to the left (or) to the right. This property of a substance of rotating the plane of polarized light is called optical activity and the substance possessing it is said to be optically active.

The observed rotation of the plane of polarized light [determined with the help of polarimeter] produced by a solution depends on :

(a) The amount of the substance in tube ;	**(b)** On the length of the sample tube;
(c) The temperature of the experiment and	**(d)** the wavelength of the light used.

The instrument used to measure angle of rotation is called polarimeter. The measurement of optical rotation is expressed in terms of specific rotation $[\alpha]_{\lambda}^{T}$; this is given by the following relation :

$$[\alpha]_{\lambda}^{T} = \frac{\alpha_{obs}}{l \times C} \quad \text{[where } \alpha = \text{observed angle of rotation]}$$

$[\alpha]_{\lambda}^{T}$ = specific rotation determined at t°C, using D-line of sodium light.

ℓ = length of solution in decimeters

C = concentration of the active compound in grams per millilitre.

The sign attached with the angle of rotation signifies the direction of rotation. Negative sign (–) indicates that the rotation is towards anti-clockwise, while positive (+) sign means that the direction of rotation is toward right clockwise.

If the substance rotates plane-polarised light to the right i.e. in clockwise direction it is called dextrorotatory and indicated by 'd' or (+)

If the substance rotates plane-polarised light to the left i.e. in anti-clockwise direction it is called laevorotatory and indicated by 'l' or (–)

Condition of Optical Activity

Compound must be asymmetrical.

Chiral Compound and Chiral Centre and asymmetric centre and pseudo chiral centre:

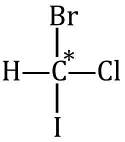
Chiral Compound:

Compound which is not super imposable on its mirror image is known as Chiral Compound. All optically active compounds are Chiral.

Asymmetric carbon (or) Chiral Carbon

If all the four bonds of carbon are satisfied by four different atoms/groups, it is chiral. Chiral carbon is designated by an asterisk (*).

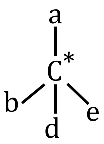
Example :



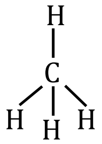
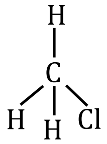
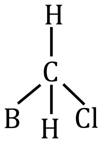
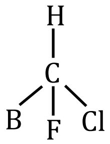
$$H-\overset{Br}{\underset{I}{\overset{*}{C}}}-Cl$$

Optically Active Carbon Compounds:

If a carbon-atom is attached with four different group, then if does not have any element of symmetry. It is known as asymmetric carbon atom, which is represented as $^*C_{abde}$.



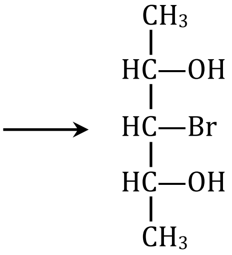
If a molecule contains only one asymmetric carbon atom, then the molecule as a whole becomes chiral and optically active and show optical isomers.

	Compound	Centre of symmetric	Plane of symmetric	Optical active
(1)		Absent	Yes	No
(2)		Absent	Yes	No
(3)		Absent	Yes	No
(4)		Absent	No	Yes

Pseudo Chiral Centre:

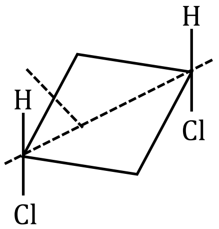
A chiral center which becomes achiral on changing the configuration of one of its substituents and vice versa

Pseudo chiral center

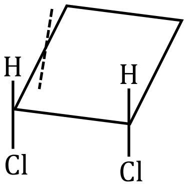

CH3
HC—OH
HC—Br
HC—OH
CH3

Symmetry Elements: Plane of Symmetry:

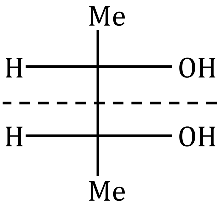
Plane of symmetry (POS) : An imaginary plane which bisects any object or molecule into two equal parts which are mirror images of each other is known as POS.



Plane of symmetry



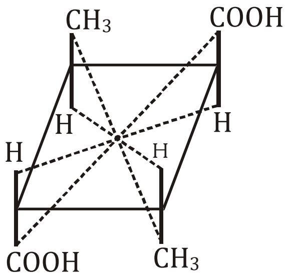
Plane of symmetry

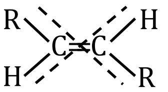

Me
H
OH
H
OH
Me

Plane of symmetry

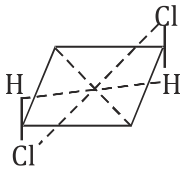
Symmetry Elements: Centre of Symmetry:

Center of Symmetry (COS): It is a point inside a molecule from which on travelling equal distance in opposite directions one takes equal time and find similar group or atom.


CH3
COOH
H
H
H
H
COOH
CH3



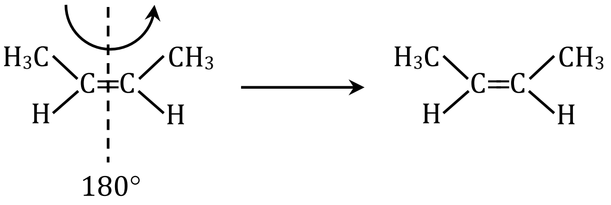
Centre of symmetry



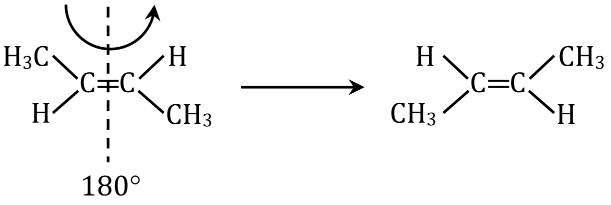
Centre of symmetry

Symmetry Elements: Axis of Symmetry, Alternate Axis of Symmetry:

(i) Axis of Symmetry (AOS) (Cn) : Axis of symmetry is an axis such that if one rotates the molecule by $\frac{360^o}{n}$, the new position of molecule is superimposable with the original one.


H3C
CH3
C=C
H
H
180°
H3C
CH3
C=C
H
H

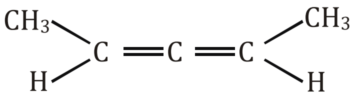
(ii) Alternate axis of Symmetry (AAOS) : An imaginary plane around which rotation by angle x gives its mirror image is known as AAOS. It is represented by S_n where n = $\frac{360}{x}$



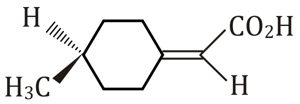
Optical isomerism in allenes, biphenyls, spiro compounds and in rigid body system:

Optically active compounds having no asymmetric carbon:

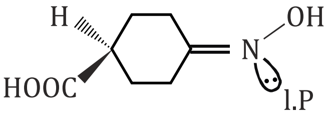
(i)



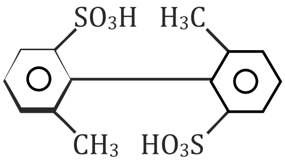
(ii)



(iii)



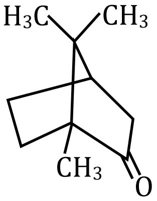
(iv)



Optically

Rigid body system:

Due to configurational restriction rigid body system has less number of total stereoisomers than expected e.g. below compound has 2 chiral centre but it exist as pair of enantiomer only.

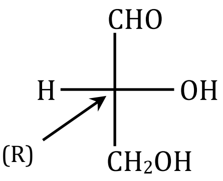


(M)

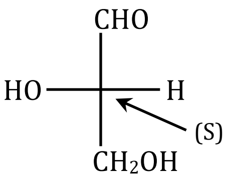
D, L Nomenclature:

D-L System (Relative configuration) : Application on correct Ficher Projection Formula:

This method is used to relate the configuration of sugars and amino acids to the enantiomers of glyceraldehyde. The configuration of (+)-glyceraldenyde has been assigned as D and the compounds with the same relative configuration are also assigned as D, & those with (-) glyceraldehyde are assigned as L.

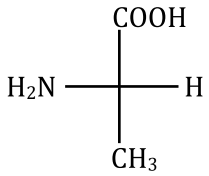


D–(+)–glyceraldehyde
–OH group on right side

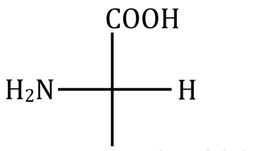


L–(–)–glyceraldehyde
–OH group on left side

Examples :

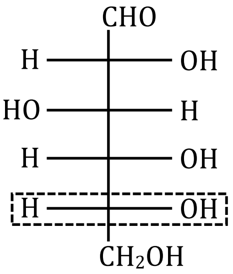


L–Alanine

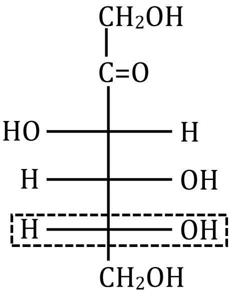


L–(+)–Glutamic acid

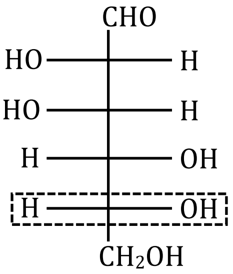
Sugars have several asymmetric carbons. A sugar whose highest numbered chiral centre (the penultimate carbon) has the same configuration as D-(+)-glyceraldehyde (–OH group on right side) is designated as a D-sugar, one whose highest numbered chiral centre has the same configuration as L-glyceraldehyde is designated as an L-sugar.



D–(+)–Glucose



D–Fructose



D–Mannose

R, S Nomenclature:

Absolute Configuration (R, S configuration)

The actual three dimensional arrangement of groups in a molecule containing asymmetric carbon is termed **absolute configuration**.

System which indicates the absolute configuration was given by three chemists R.S. Cahn, C.K. Ingold and V. Prelog. This system is known as (R) and (S) system or the Cahn–Ingold system. The letter (R) comes from the latin rectus (means right) while (S) comes from the latin sinister (means left).

(R) (S) nomenclature is assigned as follows :

Step 1 : Assign priority to the groups which are attached with chiral carbon.

Step 2 : Bring the lowest priority group to dash by even simultaneous exchanges.

Step 3 : Draw an arrow from first priority group to second priority group till third priority group.

Step 4 : If the direction of arrow is clockwise the configuration is R and if anticlockwise it is S.

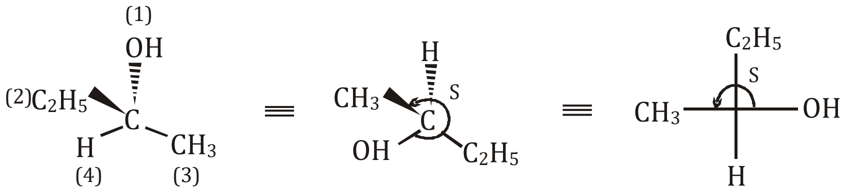
RS Nomenclature in Fischer formula :

Step 1 : Assign priority to the groups which are attached with chiral carbon.

Step 2 : Bring the lowest priority group to vertical line.

Step 3 : Draw an arrow from first priority group to second priority group till third priority group.

Step 4 : If the direction of arrow is clockwise the configuration is R and if anticlockwise it is S.



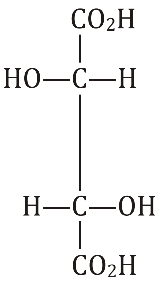
Important:

Note that the designation of a compound as R or S has nothing to do with the sign of rotation. the Cahn-Ingold rule can be applied to any three-dimensional representation of a chiral compound to determine whether it is R or S only.

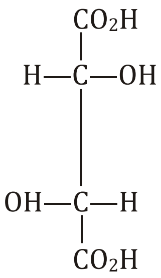
Meso Compound and Racemic Mixture:

Stereoisomerism in Tartaric Acid:

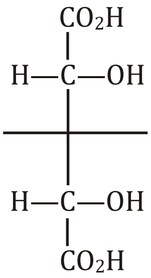
Tartaric acid exists in only three stereomeric forms. Two of them (1 and 2) are non superimposable mirror images of each other so they are enantiomers of each other. The 3rd isomer is non mirror image stereomers of (1) and (2) so it is diastereomer of (1) and (2). 3rd isomer containes plane of symmetry so, it is optically inactive. It is known as meso tartaric acid.



(1)



(2)



(3)

Meso tartaric acid

Meso isomers:

(i) Achiral compound which have chiral centers is known meso compound.

(ii) They are achiral (optical rotation = 0).

(iii) They have $[\alpha] = 0$ due to internal compensation of optical rotation.

(iv) Presence of more than one asymmetric 'C' atoms.

(v) They are non resolvable.

Internal compensation :

In meso tartaric acid the inactivity is due to effects within the molecule and not external. The force of rotation due to one of the molecules is balanced by the opposite and equal force due to the other half. The optical inactivity so produced is said to be due to internal compensation. It occurs whenever a compound containing two (or) more asymmetric carbon atoms has a plane (or) point of symmetry. Since the optical inactivity of such a compound arises within the molecule, the question of separating into active components does not arise.

Racemic mixture (External compensation):

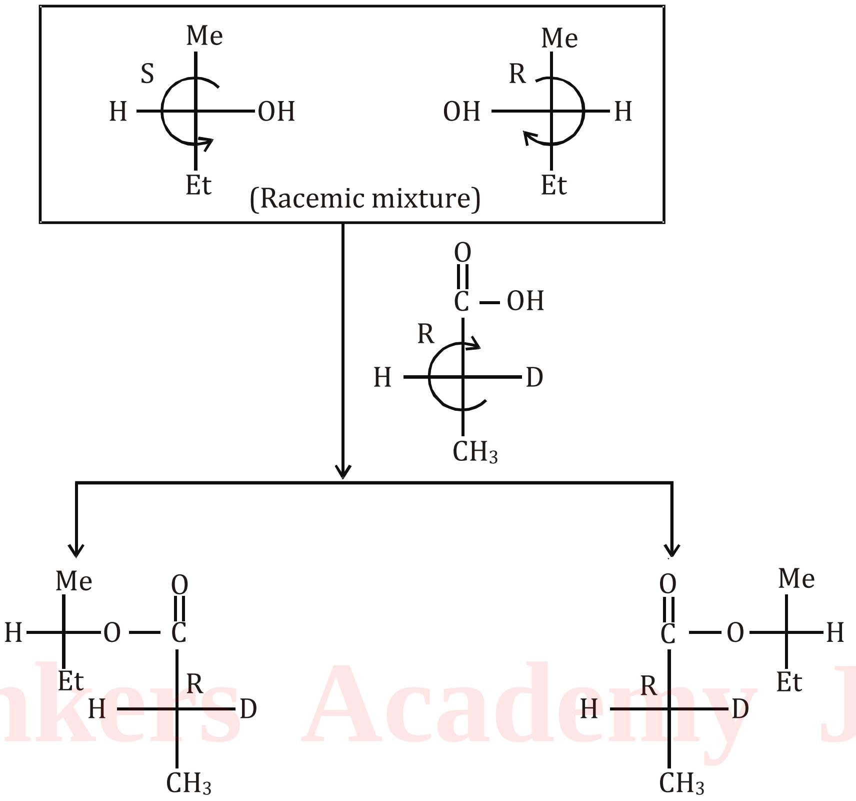
If equimolar amounts of d and ℓ-isomers are mixed in a solvent, the solution is inactive. The rotation of each isomer is balanced (or) compensated by the equal but opposite rotation of the other. Optical inactivity having this origin is described as due to external compensation. Such mixtures of (+) and (–) isomer (Racemic mixtures) can be separated into the active components.

A meso compound is optically inactive due to internal compensation.

Optical Resolution:

Resolution:- Separation of enantiomers from enantiomeric mixture is known as resolution.

When optically active compound is introducted in racemic mixture than we obtain diastereomeric pair which is easily separated by fractional distillation.



(SR) (RR)

(can be separated now because they are diastereomers)

Optical Purity and Enantiomeric Excess:

Optical Purity:

Some times we deal with mixtures that are neither optically pure nor racemic mixture. In these cases we specify the optical purity of the mixture. It is defined as the ratio of its rotation to the rotation of pure enantiomer.

$$\Rightarrow \text{Optical purity} = \frac{\text{observed optical rotation}}{\text{optical rotation of pure enantiomer}} \times 100$$

e.g. If we have some 2-butanol with observed rotation of +9.72, we compare this rotation with +13.5 rotation of the pure (+) enantiomer.

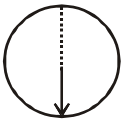
$$\text{Optical purity} = \frac{9.72}{13.5} \times 100 = 72\%$$

That means 72% is pure (+) 2-Butanol and 28% is (± mixture)

Total (+) isomer = 72 + 14 = 86%, (–) isomer = 14%

Specific Rotation(∝) :-

Rotation caused by 1gm/ml of solution in 1 dm length polarimeter tube at specific temperature and fixed wavelength of light.

Confusion  q = + 180° or – 180°

If concentration is halfed and becomes + 90° then d if – 90° then ℓ

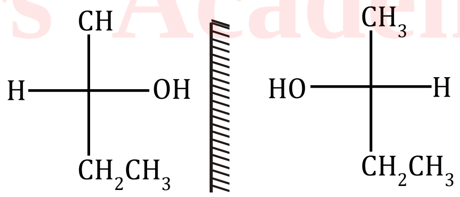
Assigning relationship in stereo isomers (Enantiomers, Diastereomers and identical isomers):

Enantiomers :

Stereoisomers which are mirror-image of each other are called enantiomers (or) enantiomorphs. All the physical and chemical properties of enantimoers are same except two :

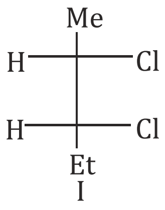
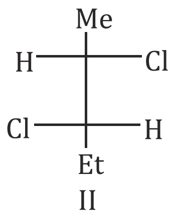
(i) They rotate PPL to the same extent but in opposite direction. One which rotates PPL in clockwise direction is called dextro-rotatory [dextro is Latin word meaning thereby right] and is designated by d(or) (+). One which rotates PPL in anti-clockwise direction is called laevo rotatory [means towards left] and designated by l (or) (–).

(ii) They react with optically active compounds with different rates.

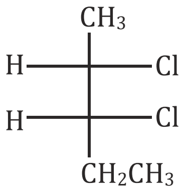


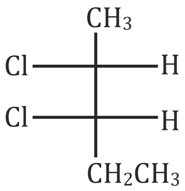
Diastereomerism :

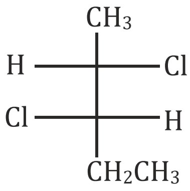
The optical isomer which are not mirror images to each other are called diastereomers.

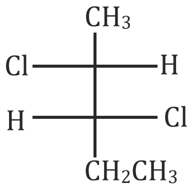
Properties	 I  II
(i) Dipole moment	Different
(ii) Melting point	Different
(iii) Boiling point	Different
(iv) Solubility & Density	Different
(v) Specific rotation	Different

Example :


(I)


(II)


(III)


(IV)

I, II = Enantiomers,

I, III = Diastereomers,

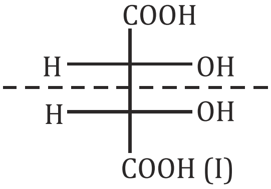
II, III = Diastereomers,

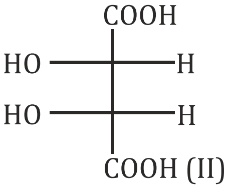
III, IV = Enantiomers

II, IV = Diastereomers

I, IV = Diastereomers

Example :

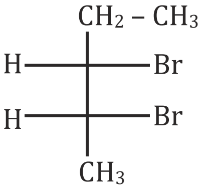

Achiral

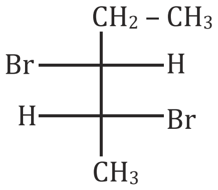


I and II are identical

Erythro and Threo:

A Diastereomer is called erythro if its Fischer projection shows similar groups on the same side of the molecule. It is called threo if similar groups are on the opposite sides of the Fischers projection.

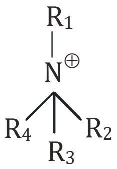

Erythro

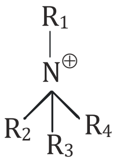

Threo

Amine Flipping:

Amine inversion:

(i) Chiral nitrogen containing tetra alkyl ammonium ion show optical isomerism.

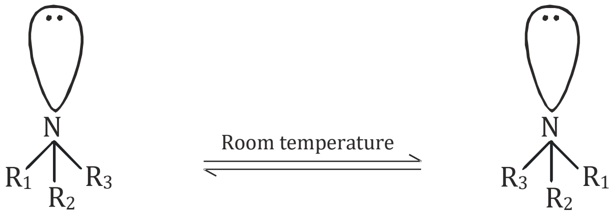

(I)


(II)

I and II enatiomers

(ii) Chiral nitrogen containing tertiary amine do not show optical isomerism

Reason :- Rapid umbrella inversion.



(I) (II)

(iii) Energy required for this interconversion is available at room temperature.

Calculation of Optical Isomers and total isomers:

Calculation of number of optical isomers:

The number of optical isomers of an organic compound depends on its structure and number of asymmetric carbon atoms. Thus, the number of optical isomers may be determined from the knowledge of the structure of the compound as follows :

(a) When the molecule is unsymmetrical

No. of optically active isomers, a = 2^n

Number of meso forms (m) = 0

$\therefore$ Total no. of optically isomers = (a + m) = 2^n

(b) When the molecule is symmetrical and has even no. of asymmetric carbon atoms.

No. of optically active isomers, a = 2^{n-1}

No. of meso forms, m = $2^{\frac{n}{2}-1}$

$\therefore$ Total no. of optically isomers = a + m

(c) When the molecule is symmetrical and has an odd no. of asymmetrical carbon atoms.

no. of optically active isomers, a = $2^{n-1} - 2^{\frac{n-1}{2}}$

No. of meso forms, m = $2^{\frac{n-1}{2}}$

$\therefore$ Total no. of optically isomer = (a + m) = 2^{n-1}